4.6.1 Additional homework

We start with some concrete problems from our undergraduate course titled "Linear Algebra: Foundations to Frontiers" [27]. If you have trouble with these, we suggest you look at Chapter 11 of that course.

Homework 4.6.1.1. Consider $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ and $b = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

- Compute an orthonormal basis for $\mathcal{C}(A)$.
- Use the method of normal equations to compute the vector $\widehat{x}$ that minimizes $\min_x \| b - Ax \|_2$
- Compute the orthogonal projection of b onto $\mathcal{C}(A)$.
- Compute the QR factorization of matrix A.
- Use the QR factorization of matrix A to compute the vector $\widehat{x}$ that minimizes $\min_x \| b - Ax \|_2$

Homework 4.6.1.2. The vectors

$$q_0 = \frac{\sqrt{2}}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}, \quad q_1 = \frac{\sqrt{2}}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}.$$

- TRUE/FALSE: These vectors are mutually orthonormal.
- Write the vector $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ as a linear combination of vectors q_0 and q_1.